



Work In Progress

Analysis of Bridges & Wormholes Paper

Part 3 – Assessment of Data Handling

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Introduction

Objective of Study

- Provide an overview of the B&W approach that makes it more accessible

Approach to Study

- Understand and explain the proposed wormhole approach to bridging
 - Control ✓
 - Data Handling
- Assess if the responsibilities of the SW Architecture proposed in the B&W paper are sufficient for the proposed approach
 - Control ✓
 - Data Handling
- Identify constraints and propose improvements
 - Control ✓
 - Data Handling

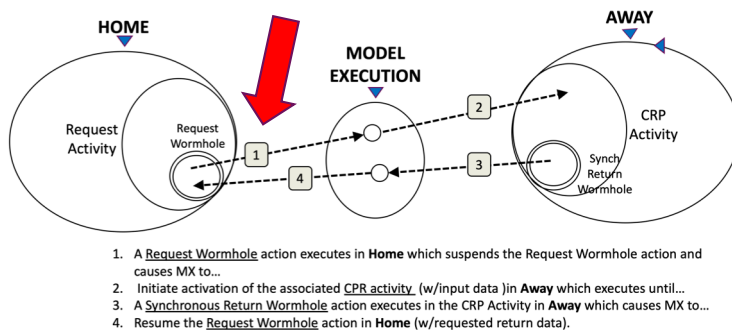


Approach to Characterizing Data Handling



Synchronous Bridge Scenario

Simplified View of a Synchronous Bridge Scenario



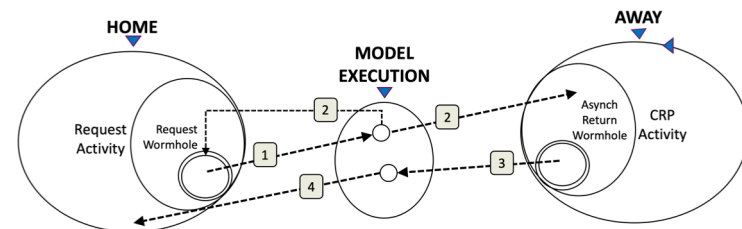
1. A Request Wormhole action executes in Home which suspends the Request Wormhole action and causes MX to...
2. Initiate activation of the associated CPR activity (w/input data) in Away which executes until...
3. A Synchronous Return Wormhole action executes in the CRP Activity in Away which causes MX to...
4. Resume the Request Wormhole action in Home (w/requested return data).

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Asynchronous Bridge Scenario

Simplified View of an Asynchronous Bridge Scenario



1. A Request Wormhole action executes in Home which suspends the Request Wormhole action and causes MX to...
2. Initiate activation of the associated CPR activity in Away and resume the Request Wormhole action in Home.
3. Eventually, an Asynchronous Return Wormhole action executes in the CRP Activity in Away which causes MX to...
4. Post the "Populated Return Vector" event in Home

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1. Use the sequencing of the Control study as a context
2. Identify the Data Handling required at each step of the sequence

Constructing the Bridge



Prescription from the Bridges and Wormholes Paper

1. Match each Wormhole with a Control Reception Point
2. Populate the participating domains
3. Develop correspondences between data values that cross the domain boundary



Matching Wormhole with CRP



Request	W1-Get Magnet <u>Current</u> (in: Magnet ID; out: current)
CRP	Read AIP (in: AIP ID; out: scaled value; return coordinates)

Diagram illustrating the mapping between the Request and the CRP for the 'Current' wormhole:

- Map Identifiers:** A dashed blue arrow points from 'Magnet ID' in the Request to 'AIP ID' in the CRP.
- Associate Values:** A dashed blue arrow points from 'current' in the Request to 'scaled value' in the CRP.

Request	W1-Get Magnet <u>Temperature</u> (in: Magnet ID; out: temperature)
CRP	Read AIP (in: AIP ID; out: scaled value; return coordinates)

Diagram illustrating the mapping between the Request and the CRP for the 'Temperature' wormhole:

- Map Identifiers:** A dashed blue arrow points from 'Magnet ID' in the Request to 'AIP ID' in the CRP.
- Associate Values:** A dashed blue arrow points from 'temperature' in the Request to 'scaled value' in the CRP.



There appear to be 3 different types of bridge data items and required handling!

1. Instance Identifier → Will need to be mapped 
2. Data Value → May need type and semantic conversions (Later)
3. Return Mechanism → Managed by MX domain (Andrew's problem)



Populate Domains

HOME

Magnet

- Magnet ID
- temperature
- current



Mag ID	Temperature	Current
1		
2		

AWAY

Analog Input Point

- AIP ID
- Register
- A
- B
- Raw value
- Scaled value



AIP ID	Scaled Value
1	
2	
3	
4	



Develop Correspondences (proposed approach)

HOME

Mag ID	Temperature	Current
1		
2		

AWAY

AIP ID	Scaled Value
1	
2	
3	
4	

Get Current ID Map

Mag ID	AIP ID
1	1
2	3

Get Temperature ID Map

Mag ID	AIP ID
1	2
2	4



Half Table Scope: Domains versus Request

Application Domain					PIO Domain		
object name ⁴	identifier attribute name	identifier attribute value	non-identifying attribute name		object name ⁴	identifier attribute name	identifier attribute value
Magnet	Magnet ID	1	current		AIP	AIP ID	1
Magnet	Magnet ID	1	current		AOP	AOP ID	1
Magnet	Magnet ID	1	temperature		AIP	AIP ID	2
Magnet	Magnet ID	2	current		AIP	AIP ID	3
Magnet	Magnet ID	2	current		AOP	AOP ID	2
Magnet	Magnet ID	2	temperature		AIP	AIP ID	4
Magnet	Magnet ID	3	current		AIP	AIP ID	5
Magnet	Magnet ID	3	current		AOP	AOP ID	3
Magnet	Magnet ID	3	temperature		AIP	AIP ID	6
...	...	...	...		...	...	...
RH Robot	Robot ID	1	x		AIP	AIP ID	96
RH Robot	Robot ID	1	x		AOP	AOP ID	17
RH Robot	Robot ID	1	y		AIP	AIP ID	97
RH Robot	Robot ID	1	y		AOP	AOP ID	18
RH Robot	Robot ID	1	θ		AIP	AIP ID	98
RH Robot	Robot ID	1	θ		AOP	AOP ID	19
...	...	...	...		...	...	...
Field probe	Field probe ID	1	x		AIP	AIP ID	114
Field probe	Field probe ID	1	x		AOP	AOP ID	20
Field probe	Field probe ID	1	y		AIP	AIP ID	115
Field probe	Field probe ID	1	y		AOP	AOP ID	21
Field probe	Field probe ID	1	z		AIP	AIP ID	116
Field probe	Field probe ID	1	z		AOP	AOP ID	21
...	...	...	...		...	...	...
Slider	Slider ID	1	position		AIP	AIP ID	48
Slider	Slider ID	1	position		AOP	AOP ID	31
...	...	...	...		...	...	...

Large and Complex

VS

Get Current ID Map

Mag ID	AIP ID
1	1
2	3

Get Temperature ID Map

Mag ID	AIP ID
1	2
2	4

Smaller and Simpler

Questions? Comments?? Suggestions???

